

# Ivan Wang

## EDUCATION

240-838-9955 • linkedin.com/in/ivanwang1 • ivanwang8989@gmail.com

### University of Maryland, College Park

Aug. 2023 – May 2027

*B.S./M.S. in Computer Science (CS Honors Program), Minors in Data Science and Philosophy*

GPA: 3.6/4.0

- **Relevant Coursework:** Algorithms, Data Structures, Machine Learning, Deep Learning, Natural Language Processing, Database Design, Operating Systems, Computer Networks, Distributed Systems, Computer Vision

## EXPERIENCE

### Software Development Engineer Intern

May 2026 – Aug. 2026

*Amazon Web Services (AWS)*

Herndon, VA

- Architected an **agentic AI system** with custom skill packages to autonomously migrate **8+ production tests** from **Scala** to **JUnit5**, boosting migration throughput **5x** for **AWS EC2 VPC** control plane APIs.
- Designed and shipped **canary infrastructure** on **AWS SDK v2**, **CDK**, and **CloudWatch** in **us-east-1**, cutting test runtime **40%** and flaky failures **60%** via **CI/CD** pipelines and **Sev2** alarms for **1M+ customers**.

### Full-Stack Software Engineer Intern

May 2025 – Sep. 2025

*University of Maryland*

College Park, MD

- Developed a full-stack web application using **Flask**, **React**, and **SQLite** to process and serve **50,000+** records, delivering **RESTful APIs** plus a responsive UI with real-time search and advanced filtering.
- Orchestrated an **ETL** pipeline in **Python** that cut processing time by **75%**, implemented schema validation and deduplication, scheduled periodic refreshes, and enabled **15+** researchers to query a live archive.

### Machine Learning Engineer Intern

Aug. 2023 – Jan. 2025

*University of Maryland*

College Park, MD

- Developed **NLP** classification models in **PyTorch**, improving text accuracy by **30%** through algorithmic optimization, hyperparameter tuning, and architecture design for higher precision and recall.
- Optimized training algorithms (SPSA, Adam) in **Python**, accelerating convergence by **25%** through optimizer and schedule tuning while reducing computational cost for scalable training.

### Software Engineer Intern

Mar. 2022 – Aug. 2023

*Washington Software*

Gaithersburg, MD

- Delivered an automated presentation generation system using **Python**, **BeautifulSoup**, and **Selenium**, reducing manual work by **40%** and saving **15+** engineering hours weekly across product teams.
- Wrote modular, reusable automation scripts across **3** product teams, cutting manual research and data-gathering time by **35%** and integrating into **CI/CD** pipelines through **Agile/Scrum** code reviews.

## PROJECTS

### Cross-Domain NLP Sentiment Classifier | *PyTorch, DistilBERT, HuggingFace*

May 2026

- Designed a **Leave-One-Domain-Out (LODO)** evaluation framework across **64,000** labeled samples from Yelp, Amazon, Twitter, and Reddit to benchmark cross-domain **NLP** generalization across **16** experimental conditions.
- Implemented a **Domain-Adversarial Neural Network (DANN)** on **DistilBERT** with gradient reversal, boosting cross-domain accuracy by **26%** over the fine-tuning baseline and achieving **74.2%** on held-out domains.

### RAG-Based Research Assistant | *LangChain, FAISS, OpenAI API, Weights & Biases*

Jan. 2025

- Built a **Retrieval-Augmented Generation (RAG)** chatbot using **LangChain** and **FAISS** to query **25+** research papers; tuned chunk size and top-*k* retrieval with **OpenAI embeddings** for accurate question answering.
- Deployed a full-stack application with **Streamlit** and integrated **Weights & Biases** for experiment tracking and hyperparameter optimization, improving reliability and achieving **95%** accuracy in policy analysis.

### ML-Driven Professor Rating Predictor | *Scikit-learn, VADER, Pandas*

Apr. 2025

- Created an end-to-end **ML pipeline** with a multithreaded web scraper to process **13,000+** professor profiles, applying **VADER** sentiment analysis for feature extraction from unstructured review data.
- Trained and evaluated regression models using **Scikit-learn** with cross-validation, comparing multiple model families, achieving **R<sup>2</sup> of 0.79** for accurate rating prediction across departments.

## TECHNICAL SKILLS

**Languages:** Python, C/C++, Java, Scala, JavaScript/TypeScript, Go, SQL, HTML/CSS, Bash, Rust, OCaml, R

**Frameworks & Libraries:** React, Flask, Node.js, PyTorch, Scikit-learn, LangChain, HuggingFace, JUnit5

**Tools & Technologies:** Git, Docker, Linux, AWS EC2/VPC, AWS SDK, CDK, CloudWatch, PostgreSQL, CI/CD

## LEADERSHIP & ACTIVITIES

**Leadership:** UMD Climbing Club Secretary (organized 8 trips/semester; increased attendance 30%); Kids For Code Vice President (delivered 400+ lessons for 3,000+ students; grew enrollment 25% through outreach and curriculum).

**Teaching & Outreach:** Panda Programmer instructor (taught Python/JS/Scratch to ages 6–12); CodeDay Technical Program Manager (secured sponsorships from **Vercel** and **Brave Browser**; led outreach to 200+ schools).